

$$(2.1) \text{ Przekształć 8.50: } \left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle = \langle \vec{x}, \vec{y} \rangle =$$

$$= [\vec{x}]_B \cdot [\vec{y}]_B$$

$$[\vec{x}]_B = [Id]_B^{K_2} [\vec{x}]_{K_2} = [Id]_B^{K_2} \cdot \vec{x}$$

$$[\vec{y}]_B = [Id]_B^{K_2} [\vec{y}]_{K_2} = [Id]_B^{K_2} \vec{y}$$

$$\rightarrow \text{Hledám } [Id]_B^{K_2}, \text{ to je: } \left(\left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} \right]_B \mid \left[\begin{pmatrix} 0 \\ 1 \end{pmatrix} \right]_B \right)$$

tedy:

$$\left(\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array} \right) \sim \left(\begin{array}{cc|cc} 1 & 0 & 7 & -3 \\ 0 & 1 & -2 & 1 \end{array} \right)$$

$$[Id]_B^{K_2} = \begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix}$$

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle = [\vec{x}]_B \cdot [\vec{y}]_B = \left([Id]_B^{K_2} \cdot \vec{x} \right) \cdot \left([Id]_B^{K_2} \cdot \vec{y} \right) =$$

$$= \left(\begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right) \cdot \left(\begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right) = \begin{pmatrix} 7x_1 - 3x_2 \\ -2x_1 + x_2 \end{pmatrix} \cdot \begin{pmatrix} 7y_1 - 3y_2 \\ -2y_1 + y_2 \end{pmatrix} =$$

$$= (7x_1 - 3x_2) \cdot (7y_1 - 3y_2) + (-2x_1 + x_2) \cdot (-2y_1 + y_2) =$$

$$= 53x_1y_1 - 23x_1y_2 - 23x_2y_1 + 10x_2y_2$$

$$\text{Matrix: } \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}^T \cdot \underbrace{\begin{pmatrix} 53 & -23 \\ -23 & 10 \end{pmatrix}}_{\text{Matrix}} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

Matrix s.s.